

SE3062 – Introduction to Artificial Intelligence & Intelligent Systems

Lecture 01: Basic Concepts, Foundations & History

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Module Overview & Assessments

Learning Outcomes:

- Explain principles of intelligent systems.
- Solve optimization problems (Linear/Non-linear).
- Design rule-based & ontology-based systems.
- Apply NLP, Fuzzy Logic, and Game Theory.

Assessments:

• Continuous (50%):

- Mid-Term (20%)
- In-lab tests ($2 \times 10\%$)
- Presentations ($2 \times 5\%$)

• Final Exam (50%)

Pass Requirement: 45% in CA and 45% in Final.

Why are we here?

- AI is no longer "Science Fiction". It is "Infrastructure".
- **Question:** How many AI systems have you used today?
 - FaceID (Computer Vision)
 - Google Maps (Search Algorithms)
 - Netflix/TikTok (Recommender Systems)
 - Email Spam Filter (Naive Bayes Classifiers)



The Big Debate: Replacement vs. Extension

The Fundamental Question

"Is AI about **replacing** humans, or **extending** human capability?"

- **Automation:** Doing boring tasks faster (e.g., Data Entry).
- **Augmentation:** Helping humans do harder tasks (e.g., AI detecting tumors doctors might miss).

Why Build Intelligent Systems? (Reasons)

Why do we need AI? It is not just about "cool tech"; it solves fundamental human constraints.

1. Overcoming Limitations

- **Speed:** AI processes data significantly faster than humans (e.g., High-frequency trading).
- **Scale:** Analyzing billions of web pages or DNA sequences is impossible for humans.
- **Safety:** Robots can operate in hazardous environments (Space, Nuclear plants, Deep sea).

2. Economic & Social Value

- **Automation:** Handling repetitive, boring tasks (Data entry).
- **Precision:** Microsurgery or manufacturing with sub-millimeter accuracy.
- **Personalization:** tailoring education or content to individual needs (Netflix/Tutors).

The Core Question

- We call ourselves *Homo sapiens* ("Man the Wise") because intelligence defines us.
- **The Goal of AI:** We don't just want to *understand* intelligence (Science); we want to *build* it (Engineering).
- **Definition:** "The study of agents that receive percepts from the environment and perform actions." (Russell & Norvig)

Fundamental Dimension of AI Definitions:

- 1 **Humanly vs. Rationally:** Do we imitate humans (imperfect) or math (ideal)?
- 2 **Thinking vs. Acting:** Do we focus on internal thought or external behavior?

Discussion: What is "Intelligence"?

Class Question

"What does 'Intelligence' mean to you?"

Discussion: What is "Intelligence"?

Common Definitions:

- **The Dictionary View (Cambridge):** "The ability to *learn, understand*, and make judgments that are based on *reason*."
- **The Psychological View:** Intelligence is a combination of three skills:
 - ① The ability to **Learn** from experience.
 - ② The ability to **Recognize** problems.
 - ③ The ability to **Solve** problems.

Transition: In AI, we must decide: Do we build intelligence that "thinks" (internal) or "acts" (external)?

The 4 Approaches to AI (The 2x2 Matrix)

Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i> , in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i> , 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

Deep Dive: The "Human" Approaches

Thinking Humanly (Cognitive Modeling)

- **Goal:** Validate AI by matching human thought traces.
- **Example:** *General Problem Solver (GPS)* (Newell & Simon, 1961). It tried to solve puzzles making the same intermediate steps (and mistakes!) as humans.
- **Field:** Cognitive Science.

Acting Humanly (Turing Test)

- **Goal:** Indistinguishable behavior.
- **Alan Turing (1950):** "Computing Machinery and Intelligence".
- **The Total Turing Test:** Requires physical interaction (Computer Vision + Robotics) to pass objects through a hatch.

Deep Dive: The "Rational" Approaches

Thinking Rationally (Laws of Thought)

- **Goal:** Irrefutable logic.
- **Origin:** Aristotle's Syllogisms ("Socrates is a man...").
- **Issue:** Hard to encode informal knowledge (uncertainty) and solve intractable problems.

Acting Rationally (Rational Agents)

- **Definition:** An agent acts to achieve the **best expected outcome**.
- **Why it wins:** It allows for reflexes (recoiling from a hot stove is rational but not "logical inference") and handles uncertainty.
- **This is the focus of this course.**

Activity 1: The "Crash Report" Tribunal (20 Mins)

Group Research & Debate

Scenario: An Autonomous Vehicle (AV) swerved to avoid a child, crossed a double yellow line (illegal), and hit a parked luxury car.

Split into Two Teams:

Team A: The Logicians

(Defending "Thinking Rationally")

Stance: The AI **failed**. It broke a strict rule (Traffic Law). AI must be logically verifiable and compliant.

Search Terms: "Rule-based AI safety", "Explainable AI compliance", "Verifiable AI".

Team B: The Pragmatists

(Defending "Acting Rationally")

Stance: The AI **succeeded**. It maximized the expected outcome (Child Life > Luxury Car). Rules are secondary to utility.

Search Terms: "MIT Moral Machine", "Utilitarian AI", "Probabilistic Robotics".

Closing the Debate: Weak AI vs. Strong AI

The debate you just had maps to the two major philosophical classifications of AI:

Weak AI (The Pragmatic View)

- Machines act *as if* they were intelligent.
- Focus: **Acting Rationally**.
- We don't care if the machine has a "mind" or "consciousness," only that it gets the job done.
- *Reality*: All current AI (Siri, ChatGPT, Tesla) is Weak AI.

Strong AI (The Philosophical View)

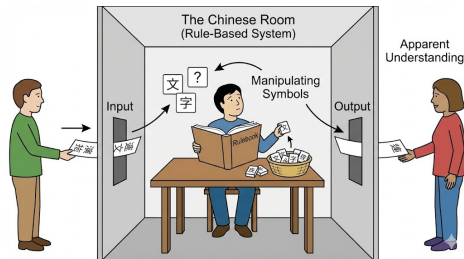
- Machines that actually *have* minds and consciousness.
- Focus: **Thinking Humanly**.
- The machine is not just a simulation; it literally understands.
- *Reality*: Does not exist yet.

Why "Acting" Wins: The Chinese Room Argument

Philosopher John Searle (1980) argued that **Acting** (Result) does not prove **Thinking** (Understanding).

- 1 A man sits in a room with a rulebook (The Program).
- 2 People pass in Chinese symbols (Input).
- 3 He looks up the symbols in the book and passes back new symbols (Output).
- 4 **Result:** He fools everyone outside into thinking he speaks Chinese (Passes Turing Test).
- 5 **Truth:** He understands *nothing*. He is just manipulating symbols.

Lesson: Simulation $\neq$ Intelligence. This is why we focus on *Acting Rationally* (Results) rather than trying to create a "Mind."



Why "Acting Rationally" Won: The Engineering Perspective

Why did the field of AI give up on imitating the human brain ("Thinking Humanly")?

1. The Argument from Aeronautics

- **Biology:** Birds fly by flapping wings.
- **Engineering:** Airplanes fly using fixed wings and jet engines.
- **Lesson:** If we insisted that airplanes must "flap" to be considered "real flying machines," we would never have invented the 747.
- *We care about **Aerodynamics** (Result), not **Biology** (Process).*

2. Measurability

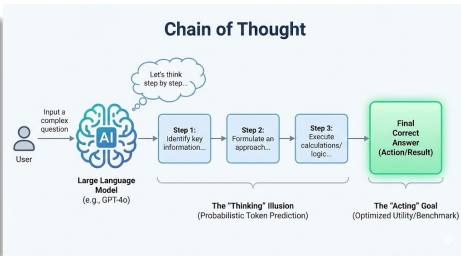
- We can measure if a car crashes (Acting). We cannot measure if the car "felt fear" (Thinking).
- Science requires testable hypotheses. "Acting" is testable; "Thinking" is philosophical.

The Modern Dilemma: Agentic & Generative AI

With the rise of Large Language Models (LLMs) and Agents, are we improving "Thinking"?

The "Chain of Thought" Illusion

- Modern AI (e.g., GPT-4o, DeepSeek) generates "reasoning steps" before answering.
- **Is this Thinking?** No. It is probabilistic token prediction.
- **Why do we do it?** Because "showing work" mathematically increases the probability of a correct final **Action**.



Conclusion: We are still "Acting"

- We judge Agentic AI by **Benchmarks** (passing exams, writing code), not by brain scans.
- We optimize for **Utility** (Results), not **Consciousness**.

The Primary Source: "Computing Machinery and Intelligence"

In 1950, Alan Turing didn't just propose the test; he predicted the arguments *against* it.

Activity: The Turing Symposium

- We will read specific sections of Turing's original paper.
- **Goal:** Critique Turing's arguments from the perspective of 2026.
- **Link:** <https://courses.cs.umbc.edu/471/papers/turing.pdf>

Activity w: The "Objections" Debate

Group Task: 1950 vs. 2026

Step 1 (Reading - 5 Mins): Your group is assigned ONE objection from Section 6. Read Turing's rebuttal.

Group Assignments:

- ① **The Theological Objection:** "Thinking is a function of man's immortal soul."
- ② **The Mathematical Objection:** "Gödel's theorem shows there are things a machine can never prove."
- ③ **The Consciousness Argument:** "Not until a machine can write a sonnet... because of thoughts and emotions felt." (Prof. Jefferson)
- ④ **Lady Lovelace's Objection:** "The Analytical Engine has no pretensions to *originate* anything."

Step 2 (Discussion - 10 Mins): Does ChatGPT/GPT-4 defeat this objection? Or does the objection still stand?

Debrief: Turing's Responses vs. Modern Reality

1. Consciousness (The "Sonnet" Argument)

- *Turing's Rebuttal*: If a machine passes a viva voce exam on a sonnet, we must accept it thinks.
- *2026 Reality*: ChatGPT can explain sonnets perfectly. Does it "feel" them? (The Chinese Room problem again).

2. Lady Lovelace (Originality)

- *Turing's Rebuttal*: Machines can "take us by surprise."
- *2026 Reality*: Hallucinations are "original" (but wrong). AlphaGo's "Move 37" was genuinely original/creative.

3. Mathematical Objection

- *Turing's Rebuttal*: Humans also make mistakes and have limits. We are not "consistent" logic machines either.

Critical Takeaway

Turing shifted the goalpost from "**Being**" (Soul/Consciousness) to "**Doing**" (Passing the Test). This defines the *Acting Humanly* paradigm.

Classification by Capability (Scope)

ANI (Narrow)

- *Artificial Narrow Intelligence*
- Good at **one** specific task.
- **Examples:** Deep Blue (Chess), Siri, Self-driving cars, Spam filters.
- *Status:* Current Reality.

AGI (General)

- *Artificial General Intelligence*
- Human-level capability across **multiple** domains.
- Can learn, plan, and create like a human.
- *Status:* Research Goal.

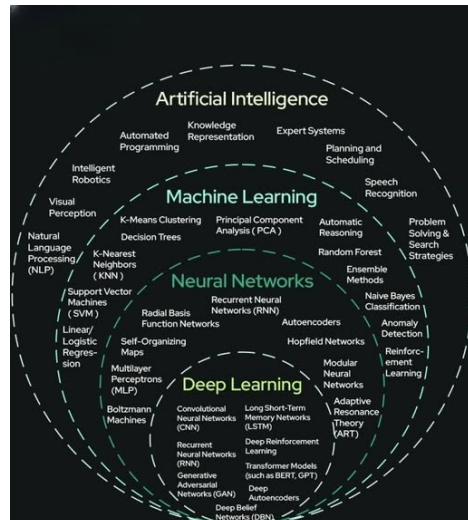
ASI (Super)

- *Artificial Super Intelligence*
- Intellect that is much smarter than the best human brains in practically every field.
- *Status:* Hypothetical / Future.

The AI Hierarchy: AI vs. ML vs. DL

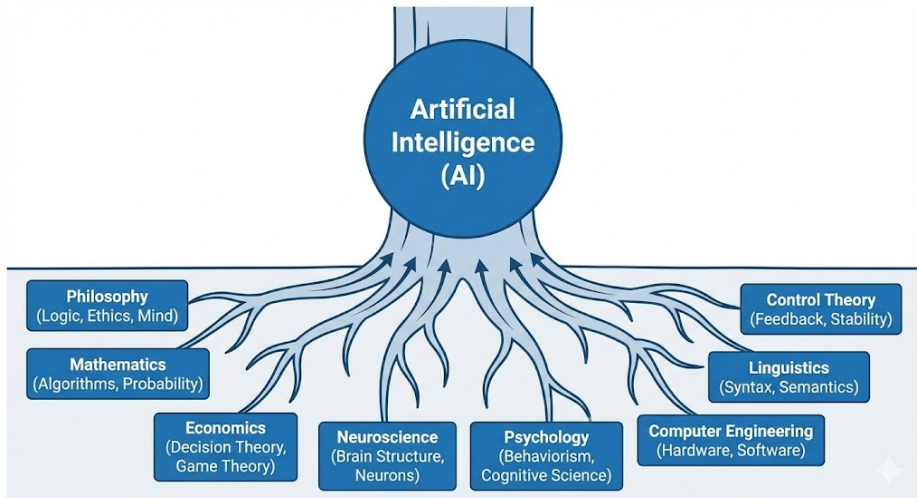
Students often confuse these terms. It is a subset relationship.

- **Artificial Intelligence (AI):** The broad umbrella. Any technique that enables computers to mimic intelligence (including Logic and Rules).
- **Machine Learning (ML):** A subset of AI. Algorithms that learn from data without explicit programming (e.g., Decision Trees, Support Vector Machines).
- **Deep Learning (DL):** A subset of ML. Uses multi-layer **Neural Networks** to solve complex problems (e.g., Image Recognition, LLMs).



Where Did AI Come From?

AI is a hybrid science. It inherits ideas from many older disciplines.



Pillar 1: Philosophy & Mathematics

Philosophy (428 B.C. – Present)

- **Logic:** Aristotle's laws of reasoning.
- **Dualism vs. Materialism:** Descartes (1600s) asked: Is the mind separate from the brain, or is it just physics?
- **Empiricism:** "Nothing is in the understanding which was not first in the senses" (John Locke).

Mathematics (c. 800 – Present)

- **Algorithms:** Al-Khowarazmi (9th Century).
- **Incompleteness:** Gödel (1931) showed that in any formal system, true statements exist that cannot be proved.
- **Intractability:** NP-Completeness (Cook Karp). Some problems are too hard to solve in reasonable time.

Pillar 2: Neuroscience, Psychology & Economics

- **Neuroscience:**

- 1861: Paul Broca proved brain localization (speech).
- 1943: McCulloch & Pitts modeled artificial neurons.
- 1949: **Hebbian Learning** ("Cells that fire together, wire together").

- **Psychology:** Behaviorism (Stimulus-Response) vs. Cognitive Psychology (Information Processing).

- **Economics: Utility Theory & Game Theory.** How to make the "best" choice when the outcome is uncertain or depends on an opponent?

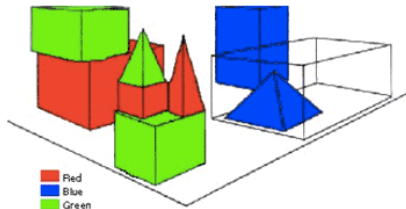
Phase 1: Gestation & Birth (1943-1956)

- **1943:** McCulloch & Pitts (First Artificial Neurons).
- **1950:** Alan Turing publishes "Computing Machinery and Intelligence".
- **1956: The Dartmouth Workshop.**
 - *Attendees:* McCarthy, Minsky, Shannon, Rochester, Newell, Simon.
 - *The Hypothesis:* "Every aspect of learning... can be so precisely described that a machine can be made to simulate it."
 - **Result:** The term "Artificial Intelligence" was coined.

Phase 2: Early Success (1952-1969)

The era of "Look, Ma, no hands!"

- **Logic Theorist (1955)**: Proved 38/52 theorems in Principia Mathematica.
- **Samuel's Checkers (1952)**: Learned to beat its creator (first Machine Learning).
- **ELIZA (1966)**: The first chatbot (simulated a psychotherapist).
- **Microworlds (SHRDLU)**: A system that could understand language to move blocks. It seemed smart but only worked in a tiny, restricted domain.



Person: Pick up a big red block.

Computer: OK.

Person: Grasp the pyramid.

Computer: I don't understand which pyramid you mean.

Phase 3: The First AI Winter (1966-1973)

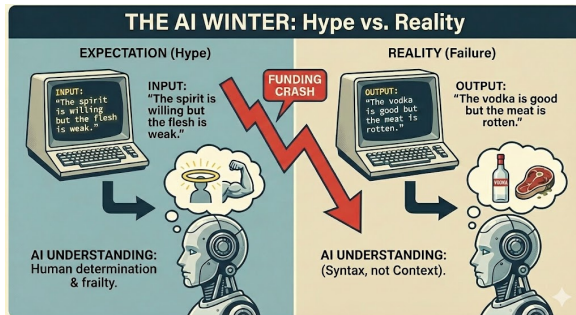
Hype met reality. Funding crashed. Why?

1 Machine Translation Failures:

- Input: "The spirit is willing but the flesh is weak."
- Output: "The vodka is good but the meat is rotten."
- Lesson: AI understood syntax but not *context*.

2 Combinatorial Explosion: Algorithms that worked for 5 blocks failed for 500.

3 Perceptrons (1969): Minsky Papert proved single-layer neural networks couldn't solve XOR problems.



Phase 4: Expert Systems (1969-1988)

The industry pivoted: "Don't be general, be a specialist."

- **Expert Systems:** Separated the *Knowledge Base* (Rules) from the *Inference Engine* (Logic).
- **Success Stories:**
 - **DENDRAL:** Chemical structure analysis.
 - **MYCIN:** Diagnosed blood infections (outperformed junior doctors).
 - **R1 (XCON):** Configured DEC computers, saving \$40M/year.
- **The Crash (Late 80s):** Expert systems were brittle. They couldn't learn and were expensive to update.

Phase 5: The Deep Learning Revolution (1986-Present)

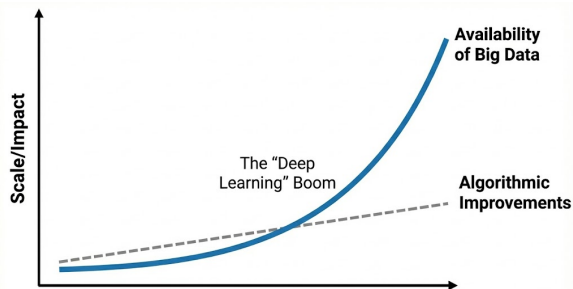
- **1986 (Backpropagation):** Rumelhart, Hinton, & Williams popularized the algorithm that allows multi-layer networks to learn.
- **1997 (Deep Blue):** IBM's system defeated Garry Kasparov in Chess.
- **2012 (AlexNet):** Deep Learning crushed the ImageNet competition. The era of GPUs and Big Data began.
- **2016 (AlphaGo):** Defeated Lee Sedol in Go (a game of intuition/pattern recognition, not just calculation).
- **2017 (Transformers):** "Attention Is All You Need" paper revolutionized NLP.
- **2022 (ChatGPT):** LLMs bring AI to the general public.

The Era of Big Data (2001-Present)

Data over Algorithms

"Simple algorithms + massive data often outperform complex methods with little data."

Application	Impact of Big Data
Word Sense Disambiguation	96% accuracy without labeled data
Image Inpainting	Photorealistic results from huge datasets
Machine Translation	Shifted entirely to data-driven models



What can AI actually do right now?

- **Robotic Vehicles:** STANLEY (won DARPA Grand Challenge 2005) paved the way for Waymo/Tesla.
- **Logistics:** DART (Dynamic Analysis and Replanning Tool) planned logistics for the 1991 Gulf War. It paid back 30 years of AI funding in a single mission.
- **Spam Fighting:** Learning algorithms classify billions of messages daily.
- **Medicine:** AI models now detect diabetic retinopathy better than ophthalmologists.

State of the Art: The Frontier (2010-2026)

We have moved from "Classification" (Is this a cat?) to "Generation" (Make a cat).

- **Scientific Discovery: AlphaFold 3** (Google DeepMind) predicts the structure of proteins, DNA, and RNA, accelerating drug discovery by years.
- **Multimodal GenAI:** Models like **GPT-4o** and **Gemini 1.5** can process video, audio, and text simultaneously in real-time.
- **Coding Assistants:** Systems like **GitHub Copilot** now write over 40% of new code in software projects.
- **Embodied AI:** Transitioning from "Chatbots" to "Physical Robots". (e.g., Tesla Optimus and Figure AI entering factory workforce).



Limitations of Current Intelligent Systems

Despite the hype, modern AI has significant flaws ("Weak AI").

- **Lack of Common Sense:** An AI can play Go, but doesn't know that "fire is hot" unless told.
- **Data Dependency:** Most systems require massive labeled datasets. They cannot learn from a single example like a child can ("One-shot learning").
- **Explainability (The "Black Box" Problem):** Deep Learning models often cannot explain *why* they made a decision (critical in Law/Medicine).
- **Bias & Ethics:** AI learns human prejudices from data (e.g., gender bias in hiring algorithms).
- **Brittleness:** Small changes in input (noise) can cause massive errors (Adversarial Attacks).

- **Definition:** We define AI as **Acting Rationally** (Agents that maximize utility).
- **Foundations:** Built on Philosophy, Math, Neuroscience, Psychology, and Economics.
- **History:** A cycle of Booms (Expert Systems, Deep Learning) and Winters (Perceptron limits).
- **Current Trend:** We are in the era of Big Data, Transformers, and Probabilistic Reasoning.

Next Week: Intelligent Agents